ASP3231 Assignment 2

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1.

The ideal diffraction limited solution refers to the maximum theoretical resolution a lens can achieve, assuming there are no lens imperfections. The only resolution arises from the physical diffraction itself. The Rayleigh criterion defines the angular resolution as

$$\theta = \frac{1.22\lambda}{D}. \tag{1}$$

Given a wavelength of 2.15 microns and a 10.5 metre diameter, $\theta = 2.498 \cdot 10^{-7}$ radians, or 51.53 milliarcseconds.

(a)

(Brown 2026)

2.

3.

4. References

Brown, M. (2026). *ASP3231 Lecture Videos*. Accessible via Moodle by staff and students of ASP3231.